

- Q.31 Enlist the instrumental errors in leveling and explain any one of them. (CO4)
- Q.32 Write a short note on reciprocal leveling. (CO4)
- Q.33 What are the advantages and disadvantages of plane table surveying? (CO5)
- Q.34 What are the various method of orientating a plane table. (CO5)
- Q.35 Define Auto level and planimeter. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Name the various accessories of the plane table. What are their functions. (CO5)
- Q.37 The following consecutive reading were taken with a level. The level was shifted after 4th, 7th and 10th readings. The first reading was taken at a B.M of R.L 100.00m. Draw out a page of level book, enter the readings and calculate the R.L of all the points by height of instrument method. 1.520, 0.965, 1.4775, 1.545, 0.435, 2.545, 1.715, 1.630, 1.915, 0.860, 1.790, 1.365 and 1.135. (CO4)
- Q.38 The following bearings were observed while traversing with a compass: (CO3)

Line	FB	BB
AB	45°45'	226°10'
BC	96°55'	227°5'
CD	29°45'	209°10'
DE	324°48'	144°48'

Mention which stations were affected by local attraction and determine the corrected bearings.

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The maximum tolerance in a 20m chain is (CO2)
a) $\pm 2\text{mm}$ b) $\pm 8\text{mm}$
c) $\pm 5\text{mm}$ d) $\pm 3\text{mm}$
- Q.2 "Three Point Problem" can be solved by (CO5)
a) Tracing Paper method b) Bessels method
c) Lehman's method d) All of the above
- Q.3 The curvature of Earth is ignored in (Co1)
a) Plane surveying b) Geodetic surveying
c) Both (a) and (b) d) None of the above
- Q.4 Survey used for determining points of strategic importance (CO1)
a) Military survey b) Mine survey
c) geological survey d) Engineering survey
- Q.5 Length of Gunter's chain is (CO2)
a) 20 m b) 30 m
c) 66 ft d) 100 ft
- Q.6 Which of the following is an obstacle to chaining but not to ranging? (CO2)
a) Hillock b) River
c) Building d) None of these

Q.7 In reduced bearing system, bearing is measured from: (CO3)

- a) Nearest one (North or south)
- b) South
- c) North
- d) West

Q.8 The following sights are taken on a "Turning point" (CO4)

- a) Back sight only
- b) Foresight only
- c) Foresight and Backsight
- d) Foresight and Intermediate sight

Q.9 The instrument used for accurate centering in plane table survey is (CO5)

- a) Spirit level
- b) Plumbing fork (U-fork)
- c) Alidade
- d) Trough compass

Q.10 A relatively fixed point of known elevation above datum, is called (CO4)

- a) Datum point
- b) Reduced level
- c) Bench mark
- d) Reference point

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The sum of the interior angles of a closed traverse is _____ (CO3)

Q.12 Intersection method is used for locating the position of objects. (CO5)

Q.13 The errors which occurs in _____ directions are called compensating error. (CO2)

Q.14 _____ compass can be used without fixing it on tripod stand. (CO3)

Q.15 Define angular measurement. (CO1)

Q.16 Define principle of chain surveying. (CO2)

Q.17 Define axis of telescope. (CO4)

Q.18 Define a traverse. (CO5)

Q.19 Write the use of alidade in plane table surveying (CO5)

Q.20 The size of plane table is. (600 mm X 750 mm / 300 mm x 450 mm) (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are the various method of ORIENTATION a plane table. (CO5)

Q.22 Explain in brief the principles of surveying. (CO1)

Q.23 What are the causes of errors in chain surveying? (CO2)

Q.24 Enlist the natural errors in leveling and explain any one of them. (CO4)

Q.25 Explain the concept of two-point problem in plane table surveying. (CO5)

Q.26 Name the different classification of surveying based on nature of field survey and describe any one of them. (CO1)

Q.27 What are the obstacles in chain surveying? (CO2)

Q.28 Convert the following QB to WCB: (Co3)

- a) N22°30'E
- b) S9°48'E
- c) S31°54'W
- d) N32°36'W
- e) N5°42'W

Q.29 Convert the following WCB to QB: (CO3)

- a) 142°30'
- b) 223°15'
- c) 60°10'
- d) 287°15'
- e) 12°45'

Q.30 Find the correction for curvature and refraction for distance of (CO4)

- a) 2933 meters
- b) 7Km